Apply filters to SQL queries

Project description

In this project, I used SQL to investigate potential security issues involving employee login activity and company devices. I queried and filtered data from the `log_in_attempts` and `employees` tables to identify suspicious login activity and determine which employees needed security updates. I used operators such as `AND`, `OR`, `NOT`, and `LIKE` to narrow the results to the specific information needed for each investigation.

Retrieve after hours failed login attempts

I queried the `log_in_attempts` table to investigate failed login attempts that occurred after normal business hours. I used `login_time > '18:00'` to return attempts after 6:00 PM and `success = 0` to identify unsuccessful login attempts. I used the `AND` operator because both conditions had to be true. The query returned 19 failed after-hours login attempts.

Retrieve login attempts on specific dates

I queried the `log_in_attempts` table to review login activity from May 8 and May 9, 2022. I used the `WHERE` clause to filter the `login_date` column and the `OR` operator because a record could match either date. The query returned 75 login attempts from the two dates being investigated.

Retrieve login attempts outside of Mexico

I queried the `log_in_attempts` table to investigate login attempts that did not originate in Mexico. I used `NOT` with `LIKE 'MEX%'` to exclude country values beginning with `MEX`. The `%` wildcard allows the pattern to match both `MEX` and `MEXICO`. This query returned 144 login attempts that originated outside of Mexico.

Retrieve employees in Marketing

I queried the `employees` table to identify employees in the Marketing department who work in the East building. I used `department = 'Marketing'` to filter by department and `office LIKE 'East%'` to find offices that begin with East. I used the `AND` operator because both conditions had to be true. The query returned 7 employees who met both conditions.

Retrieve employees in Finance or Sales

I queried the `employees` table to identify employees who work in either the Finance or Sales department. I used the `OR` operator because an employee only needed to meet one of the two conditions. The query returned 71 employees from the Finance and Sales departments.

Retrieve all employees not in IT

I queried the `employees` table to identify employees who are not in the Information Technology department. I used the `NOT` operator with `department = 'Information Technology'` to exclude IT employees from the results. This query returned 161 employees from the other departments who still need the security update.

Summary

I used SQL queries to investigate potential security issues involving login attempts and employee machines. I filtered data using `AND`, `OR`, `NOT`, and `LIKE` to identify failed after-hours logins, activity on specific dates, login attempts outside of Mexico, and employees who needed security updates. These queries helped narrow down large amounts of data so I could focus on information relevant to the security investigation.